V2.0.6

2024.10

GCU

私有通信协议

阅读提示 - 符号说明

重要注意事项	操作提示	词汇解释及参考信息

版本历史

日期	文档版本	协议版本
2023.06.19	V2.0	-

日期	文档版本	协议版本
2023.08.09	V2.0.1	V0.0

1. 串口通信波特率改为自适应。网络通信增加 TCP Server 模式。[P1]
2. 协议内容增加版本号，修正 GCU 返回数据包协议头的错误。[P2]
3. 上位机发送数据包主帧：
   3.1. 滚转 / 俯仰 / 指向控制量（5~10 字节）增加期望欧拉角与期望相对角；[P3]
   3.2. 状态标志（11 字节）增加控制量有效性标识（B2 位）。[P3]
4. 上位机发送数据包副帧：
   4.1. 删去载机距离 home 点距离（57~60 字节）；[P4]
   4.2. 增加相对高度（57~60 字节）。[P4]
5. GCU 返回数据包主帧：
   5.1. 吊舱工作模式（5 字节）增加 FPV 模式与欧拉角控制模式；[P5]
   5.2. 相机状态（6~7 字节）删去光圈快门状态（B11 位）。[P5]
6. GCU 返回数据包副帧：
   6.1. 删去 59~61 字节内容；[P6]
   6.2. 增加 1 号相机当前倍率(59~60 字节)与 2 号相机当前倍率(61~62 字节)。[P6]
7. 控制命令与命令反馈：
   7.1. 增加空命令的说明；[P7]
   7.2. 增加 FPV 模式命令、欧拉角控制模式命令、外部图像跟踪模式命令及 OSD 命令；[P7~P9]
   7.3. 详细了凝视模式的说明文字；[P7]
   7.4. 修改了拍照、录像、聚焦、调色盘及夜视命令的参数。[P8~P9]
8. 更新了示例数据包。[P11~P16]

日期	文档版本	协议版本
2023.10.12	V2.0.2	V0.1

1. 增加协议字节序的说明。[P2]
2. 上位机发送数据包主帧：
   2.1 载机绝对滚转、俯仰、偏航角（12~17 字节）说明中增加坐标系定义。[P3]
3. GCU 返回数据包主帧：
   3.1 相机状态标志（6~7 字节）增加补光灯状态（B10 位）。[P5]
   3.2 修正垂直方向脱靶量（10~11 字节）坐标轴方向的错误（“向上为正”→“向下为正”）。[P5]
   3.3 水平 / 垂直方向脱靶量（8~11 字节）说明中增加数值范围。[P5]
   3.4 相机绝对角速度（24~29 字节）说明中增加坐标系定义及旋转顺序。[P5]
4. GCU 返回数据包副帧：
   4.1 修正了目标纬度(51~54 字节)分辨率单位的错误(“1mm”→“1e-7deg”)。[P6]
5. 控制命令与命令反馈：
   5.1 修改了 FPV 模式、锁定模式及跟随模式说明中的控制量描述。[P7]
   5.2 修正了凝视模式（地理坐标引导）失败命令反馈的错误（“0x015 0x01”→“0x15 0x01”）。[P7]
   5.3 跟踪模式说明中增加目标框左上角及右下角坐标。[P8]
   5.4 指点平移模式说明中增加画面左上角及右下角坐标。[P8]
   5.5 外部图像跟踪模式说明中增加画面中心、左上角及右下角的脱靶量。[P8]
6. 增加附录 1：上位机发送数据包转换示例。[P11]
7. 增加附录 2：载机坐标系定义。[P12]
8. 增加附录 3：相机坐标系定义及旋转顺序。[P13]
9. 增加附录 5：GPS 时间与 UTC 转换函数。[P20]

日期	文档版本	协议版本
2024.06.14	V2.0.5	V0.2

1. 上位机发送数据包主帧：
   1.1 状态标志说明（11 字节）中增加关于控制量有效性的解释（B2 位）。[P3]
2. GCU 返回数据包主帧：
   2.1 吊舱工作模式（5 字节）中，将 FPV 模式更名为角度控制模式 1，增加角度控制模式 2。[P5]
   2.2 将相机状态标志（6~7 字节）更名为吊舱状态标志。[P5]
3. GCU 返回数据包副帧：
   3.1 更新故障代码（41~42 字节）。[P6]
   3.2 增加热成像相机状态标志（63 字节）。[P6]
   3.3 增加相机状态标志（64~65 字节）。[P6]
   3.4 增加时区（66 字节）。[P6]
4. 控制命令与命令反馈：
   4.1 增加 OSD 坐标设置命令、图像自动翻转命令与时区设置命令。[P7]
   4.2 修改了角度控制模式 1（原 FPV 模式）命令、跟随模式命令及欧拉角控制模式命令的说明。[P7]
   4.3 增加角度控制模式 2 命令。[P8]
   4.4 修改了调色盘命令的参数范围（[0, 100]->[0, 10]）。[P9]
   4.5 增加区域测温命令、温度报警命令、等温线命令及指点测温命令。[P10]
   4.6 画中画命令增加了切换至指定模式的功能。[P10]
   4.7 增加了目标识别命令与变焦相机数字变焦命令。[P10]
5. 增加附录 2：GCU 返回数据包转换示例。[P13]
6. 更新附录 5：示例数据包。[P17]
7. 增加附录 7：吊舱型号代码。[P27]

日期	文档版本	协议版本
2024.10.24	V2.0.6	V0.2

1. GCU 返回数据主帧：
   1.1 吊舱工作模式（5 字节）中，将角度控制模式 1 更名为角度控制模式，将锁定模式更名为指向锁定模式，将跟随模式更名为指向跟随模式，将角度控制模式 2 更名为 FPV 模式。[P4]
2. 控制命令与命令反馈：
   2.1 将命令反馈拆分为指令 + 执行状态。[P8]
   2.2 修改了吊舱功能相关命令的说明。[P8]
3. 更新附录 7：吊舱型号代码。[P28]
4. 增加附录 8：控制命令支持列表。[P29]

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接口配置

串口配置

- 串口电平：TTL
- 数据位：8
- 停止位：1
- 校验位：无
- 通信模式：全双工
- 波特率：支持 115200、250000、500000 和 1000000。
- 通信频率：建议为 30-50Hz，频率越高，控制效果越好。通信频率不得过低或停发数据。

网络配置

- UDP 模式：发送源端口为 2337，目的地址默认为局域网内广播地址，目的端口为 2338。
- TCP Server 模式：对端需设置为 TCP Client 模式，远程 IP 地址与 GCU 一致，远程端口号为 2332。

总述

协议通信逻辑为问答模式，上位机首先发送数据包，GCU 接收到正确的数据包后会发送返回数据包。一个完整的数据包由协议头、包长度、数据主帧、数据副帧、控制命令或命令反馈及 CRC 校验数据组成。数据包总长度为 S 字节。其中控制命令或命令反馈部分长度可变。

控制命令或反馈包含指令和参数 / 执行状态两部分，不同的指令对应的参数不同，详见本文档《数据帧》部分。如连续发送带有相同指令的控制命令（即使参数不同），GCU 只会执行一次操作，如要多次触发相同的功能，需要使用含有空命令的数据包隔开。

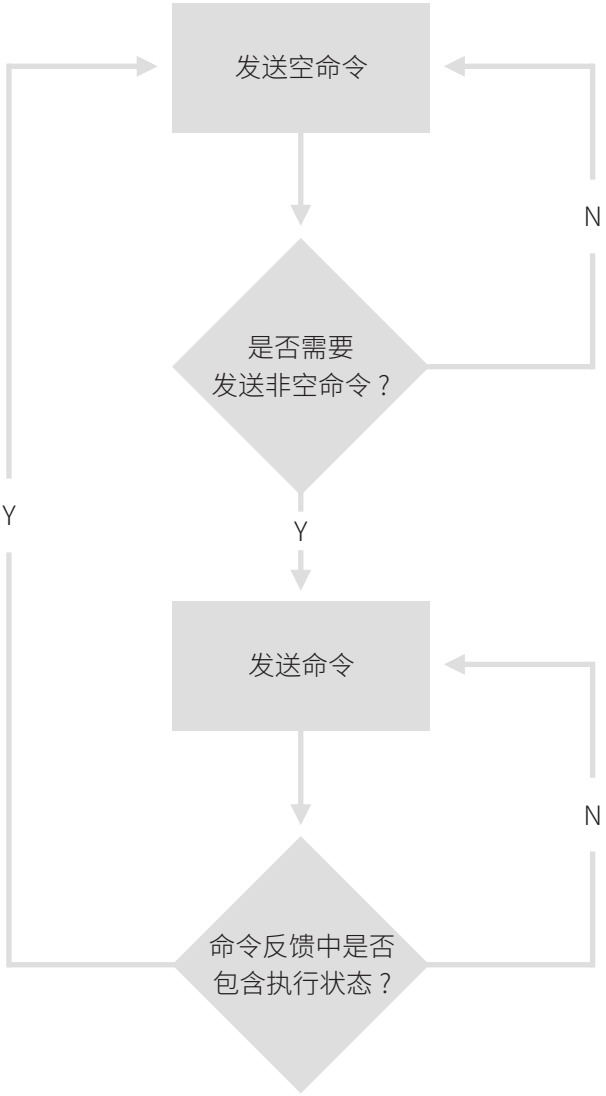
数据包结构如下表所示。

上位机发送数据包结构									
0	1	2~3	4	5~36	37~68	69	70~S-3	S-2	S-1
0xA8	0xE5	包长度	版本号	数据主帧	数据副帧	指令	参数	CRC 高	CRC 低
		U16	U8	32 字节	32 字节	U8	长度可变	U16	
GCU 返回数据包结构									
0	1	2~3	4	5~36	37~68	69	70~S-3	S-2	S-1
0x8A	0x5E	包长度	版本号	数据主帧	数据副帧	指令	执行状态	CRC 高	CRC 低
		U16	U8	32 字节	32 字节	U8	长度可变	U16	

CRC 校验内容为 0~S-3 字节数据。

本协议采用小端序（CRC 除外）。

无参数时上位机发送数据包中无 70~S-3 字节；无执行状态时 GCU 返回数据包中无 70~S-3 字节。


发送空命令
N
是否需要
发送非空命令？
Y
Y
发送命令
N
命令反馈中是否
包含执行状态？

数据帧

上位机发送数据主帧格式

<table>
<tr><th>字节</th><th>内容</th><th colspan="2">说明</th></tr>
<tr><td>5~6</td><td>滚转控制量，仅在角度控制模式及欧拉角控制模式生效，其他模式下此值无效</td><td colspan="2" rowspan="3">S16。
控制量为期望角速率时，数据范围 [-1500,1500]，分辨率为 0.1/ 当前主画面倍数（° /s）；
控制量为期望欧拉角时，数据范围 [-18000,18000]，分辨率 0.01°；
控制量为吊舱与载机的期望相对角度时，数据范围 [-18000,18000]，分辨率 0.01°</td></tr>
<tr><td>7~8</td><td>俯仰控制量</td></tr>
<tr><td>9~10</td><td>偏航控制量</td></tr>
<tr><td>11</td><td>状态标志</td><td colspan="2">B7~B3：预留 , 此位为 0
B2：0- 控制量无效，此时期望角速率为 0，期望欧拉角与期望相对角度维持当前值；1- 控制量有效
B1：预留 , 此位为 0
B0：0- 载机惯导数据无效；1- 载机惯导数据有效</td></tr>
<tr><td>12~13</td><td>载机绝对滚转角</td><td>S16，[-18000,18000），分辨率 0.01deg</td><td rowspan="3">载机坐标系定义详见附录 2</td></tr>
<tr><td>14~15</td><td>载机绝对俯仰角</td><td>S16，[-9000,9000]，分辨率 0.01deg</td></tr>
<tr><td>16~17</td><td>载机绝对偏航角</td><td>U16，[0,36000），分辨率 0.01deg</td></tr>
<tr><td>18~19</td><td>载机北向加速度</td><td>S16，分辨率 0.01m/s^2，北向为正</td><td rowspan="3">载机静止或匀速直线飞行时，三个加速度均为零</td></tr>
<tr><td>20~21</td><td>载机东向加速度</td><td>S16，分辨率 0.01m/s^2，东向为正</td></tr>
<tr><td>22~23</td><td>载机天向加速度</td><td>S16，分辨率 0.01m/s^2，天向为正</td></tr>
<tr><td>24~25</td><td>载机北向速度</td><td colspan="2">S16，分辨率 0.01m/s，北向为正</td></tr>
<tr><td>26~27</td><td>载机东向速度</td><td colspan="2">S16，分辨率 0.01m/s，东向为正</td></tr>
<tr><td>28~29</td><td>载机天向速度</td><td colspan="2">S16，分辨率 0.01m/s，天向为正</td></tr>
<tr><td>30</td><td>GCU 返回数据副帧请求码</td><td colspan="2">填写需要返回的副帧帧头，如不需要请求副帧，此处为 0x00</td></tr>
<tr><td>31~36</td><td>预留</td><td colspan="2">0x00</td></tr>
</table>

12~29 位数据非常重要，不正确的数据会导致吊舱姿态计算错误。

上位机发送数据副帧格式

字节	内容	说明
37	0x01	副帧帧头
38~41	载机经度	S32，分辨率 1e-7deg
42~45	载机纬度	S32，分辨率 1e-7deg
46~49	载机海拔高度	S32，分辨率 1mm
50	载机卫星数	U8
51~54	载机 GNSS 毫秒数	U32
55~56	载机 GNSS 周数	S16
57~60	相对高度	S32，分辨率 1mm，如不需要可为 0
61~68	预留	0x00

如无副帧数据，则 37~68 字节均为 0x00。

GCU 返回数据主帧格式

字节	内容	说明
5	吊舱工作模式	0x10- 角度控制模式； 0x11- 指向锁定模式； 0x12- 指向跟随模式； 0x13- 俯拍模式； 0x14- 欧拉角控制模式； 0x16- 凝视模式； 0x17- 跟踪模式； 0x1C- FPV 模式
6~7	吊舱状态标志	B15~B13：预留 B12：0- 正置上电；1- 倒置上电 B11：预留 B10：0- 补光关；1- 补光开 B9：0- 夜视关；1- 夜视开 B8：0- 测距关；1- 测距开 B7：0- 测距值与目标坐标无效； 1- 测距值与目标坐标有效 B6~B1：预留 B0：0- 跟踪目标失败，脱靶量数据无效； 1- 跟踪目标成功，脱靶量数据有效
8~9	水平方向脱靶量	S16，[-1000,1000]，原点为图像中心，向右为正
10~11	垂直方向脱靶量	S16，[-1000,1000]，原点为图像中心，向下为正
12~13	相机 X 轴相对角度	S16，[-18000,18000)，相机相对于载机的角度值，分辨率 0.01deg。此数据由电机编码器角度解算得出，不依赖载机惯导数据。相机坐标系定义及旋转顺序详见附录 3
14~15	相机 Y 轴相对角度	
16~17	相机 Z 轴相对角度	
18~19	相机绝对滚转角	S16，[-9000,9000] 欧拉角，分辨率 0.01deg。此数据依赖载机惯导数据
20~21	相机绝对俯仰角	S16，[-18000,18000)
22~23	相机绝对偏航角	U16，[0,36000)
24~25	相机 X 轴绝对角速度	S16, 分辨率 0.1deg/s，相机坐标系定义及旋转顺序详见附录 3
26~27	相机 Y 轴绝对角速度	
28~29	相机 Z 轴绝对角速度	
30~36	预留	-

? 脱靶量：目标于画面中所处的位置与画面中心的偏移量。

GCU 返回数据副帧格式

字节	内容	说明	
37	0x01	副帧帧头	
38	硬件版本	U8	
39	固件版本	U8	
40	吊舱代码	吊舱型号代码详见附录 7	
41~42	故障代码	B15：GCU 硬件故障； B14：GNSS 未定位； B13：Mavlink 通信频率异常； B12~B8：预留； B7：吊舱硬件故障； B6~B0：预留	
43~46	目标距离	S32，测距仪测得的目标距离，分辨率 0.1m 输出 -1m 或 0 时表示距离无效	
47~50	目标经度	S32，分辨率 1e-7deg	此数据依赖载机惯导数据
51~54	目标纬度	S32，分辨率 1e-7deg	
55~58	目标海拔高度	S32，分辨率 1mm	
59~60	1 号相机（默认为可见光变焦相机）当前倍率	U16，分辨率 0.1x	
61~62	2 号相机（默认为热成像相机）当前倍率	U16，分辨率 0.1x	
63	热成像相机状态标志	B7：0- 测温不可用，1- 测温可用； B6：0- 区域测温关，1- 区域测温开； B5：0- 温度报警关，1- 温度报警开； B4：0- 等温线关，1- 等温线开； B3：0- 指点测温关，1- 指点测温开； B2：预留； B1：高温报警； B0：低温报警	
64~65	相机状态标志	B15：0- 目标检测关，1- 目标检测开； B14：0- 数字变倍关，1- 数字变倍开； B13：0-OSD 显示关，1-OSD 显示开； B12：0-OSD 载机坐标，1-OSD 目标点坐标； B11：0- 图像自动翻转开，1- 图像自动翻转关； B10~B5：预留； B4：0- 未录像，1- 录像中； B3：预留； B2~B0：uint_t，画中画模式	
66	时区	S8	
67~68	预留	-	

当请求的副帧头非法时，37~68 字节均为 0x00。

控制命令与命令反馈

空命令

功能	指令	参数	执行状态	说明
空命令	0x00	-	-	用于隔开带有相同指令的控制命令

吊舱功能

功能	指令	参数	执行状态	说明
吊舱校准	0x01	-	成功：0x00 失败：0x01 执行中：0x02	校准时需保持吊舱静止，校准过程将持续数秒钟
回中	0x03	-	成功：0x00 失败：0x01	锁定与跟随模式下，吊舱俯仰及指向回中，工作模式不变； 俯拍模式下，吊舱指向回中，工作模式不变； 角度控制、欧拉角控制、凝视模式、跟踪模式及 FPV 模式下，吊舱不响应回中命令； 执行过程中，控制量需置为无效
角度控制	0x10	-	成功：0x00 失败：0x01	控制量： 滚转 - 期望欧拉角； 俯仰 - 期望欧拉角； 偏航 - 期望相对角度
指向锁定	0x11	-	成功：0x00 失败：0x01	控制量： 滚转 - 无效； 俯仰 - 期望角速度； 偏航 - 期望角速度
指向跟随	0x12	-	成功：0x00 失败：0x01	控制量： 滚转 - 无效； 俯仰 - 期望角速度； 偏航 - 期望角速度（为 0 或无效时跟随载机）
俯拍	0x13	-	成功：0x00 失败：0x01	控制量： 滚转 - 无效； 俯仰 - 无效（欧拉角 -90°）； 偏航 - 期望角速度（由指向跟随模式切换到俯拍模式后，控制量为 0 或无效时跟随载机）
欧拉角控制	0x14	-	成功：0x00 失败：0x01	控制量： 滚转 - 期望欧拉角； 俯仰 - 期望欧拉角； 偏航 - 期望欧拉角

功能	指令	参数	执行状态	说明
凝视（地理坐标引导）	0x15	PP PP PP PP QQ QQ QQ QQ RR RR RR RR	成功：0x00 失败：0x01	控制量为 0 或无效时，吊舱锁定目标地理坐标点； PP PP PP PP / QQ QQ QQ QQ / RR RR RR RR 分别为目标点经度、纬度与海拔高度； PP PP PP PP：S32，分辨率 1e-7deg； QQ QQ QQ QQ：S32，分辨率 1e-7 deg； RR RR RR RR：S32，分辨率 1mm； 实现本功能时，需保证 GCU 收到有效载机惯导数据； 控制量不为 0 且有效时，吊舱将切换至凝视（地理目标锁定）
凝视（地理目标锁定）	0x16	-	成功：0x00 失败：0x01	控制量为 0 或无效时，吊舱锁定画面中心地理目标； 控制量： 滚转 - 无效； 俯仰 - 期望角速度； 偏航 - 期望角速度； 实现本功能时，需保证 GCU 收到有效载机惯导数据
跟踪	0x17	0x01 TT X0 X0 Y0 Y0 X1 X1 Y1 Y1	成功：0x00 失败：0x01	TT : U8，0x01- 进入跟踪； 0x00- 退出跟踪 X0 X0 / Y0 Y0 / X1 X1 / Y1 Y1： U16，[0,10000]，分别为目标框左上角与右下角的水平与垂直坐标，原点为图像左上角，X 坐标向右为正，Y 坐标向下为正，左上角坐标为 [0,0], 右下角坐标为 [10000,10000] 此功能通过吊舱内置图像处理实现； 控制量无效
指点平移	0x1A	0x01 X0 X0 Y0 Y0	成功：0x00 失败：0x01	X0 X0 / Y0 Y0：U16，[0,10000]，分别为目标点水平与垂直坐标，原点为图像左上角，X 轴向右为正，Y 轴向下为正，左上角坐标为 [0,0], 右下角坐标为 [10000,10000]；执行过程中，控制量需置为无效。此命令执行后，吊舱将切换为指向锁定模式
FPV	0x1C	-	成功：0x00 失败：0x01	控制量： 滚转 - 期望相对角度； 俯仰 - 期望相对角度； 偏航 - 期望相对角度

相机功能

功能	指令	参数	执行状态	说明
OSD 坐标	0x06	TT	成功：0x00 失败：0x01	TT：U8，0x00- 载机坐标； 0x01- 目标坐标
图像自动翻转	0x07	TT	成功：0x00 失败：0x01	TT：U8，0x00- 自动翻转开； 0x01- 自动翻转关
时区	0x08	TT	成功：0x00 失败：0x01	TT：S8
拍照	0x20	0x01	成功：0x00 失败：0x01	-
录像开始 / 停止	0x21	0x01	成功：0x00 失败：0x01	-
连续放大	0x22	KK	成功：0x00 失败：NN	-
连续缩小	0x23	KK	成功：0x00 失败：NN	-
停止变倍	0x24	KK	成功：0x00 失败：NN	-
指定倍率	0x25	KK ZZ ZZ	成功：0x00 失败：NN	ZZ ZZ：S16，[-32768, -10]，[1, 10000]，负值区为期望倍率，分辨率为 0.1x；正值区为期望倍率比例，1 对应最小倍率，10000 对应最大倍率。本命令对应的最大倍率以吊舱所能实现为准 以最大倍率为 30 倍的相机为例，-10 与 1 均对应 1 倍，-150 与 5000 均对应 15 倍，-300 与 10000 均对应 30 倍
聚焦	0x26	0x01	成功：0x00 失败：0x01	-
调色盘	0x2A	0x02 TT	成功：0x00 失败：0x02	TT： U8，[0,10]，要切换的热成像调色盘模式。此值为 0 时，调色盘循环切换
夜视	0x2B	0x01 TT	成功：0x00 失败：0x01	TT : U8，0x00- 夜视关； 0x01- 夜视开； 0x02- 自动

KK 表示触发执行的相机序号，NN 表示执行失败的相机序号。KK 与 NN 数据类型均为 U8，B7~B0 对应 8 号相机 ~1 号相机，某位为 1 表示该位对应的相机被标记。例如 0x03（00000011）表示 1 号相机与 2 号相机。1 号相机默认为可见光变焦相机，2 号相机默认为热成像相机。

功能	指令	参数	执行状态	说明
区域测温	0x30	0x02 TT X0 X0 Y0 Y0 X1 X1 Y1 Y1	成功：0x00 失败：0x02	TT：U8，0x00- 关闭区域测温； 0x01- 开启区域测温 X0 X0 / Y0 Y0 / X1 X1 / Y1 Y1：U16，[0,10000]，分别为测温区域框左上角与右下角 的水平与垂直坐标，原点为图像左上角，X 坐标向右为正，Y 坐标向下为正，左上角坐 标为 [0,0]，右下角坐标为 [10000,10000]
温度报警	0x31	0x02 TT HH HH LL LL	成功：0x00 失败：0x02	TT：U8，0x00- 关闭温度报警； 0x01- 开启温度报警 HH HH / LL LL：S16，高低温警告温度，分辨率 0.1°C
等温线	0x32	0x02 TT HH HH LL LL	成功：0x00 失败：0x02	TT：U8，0x00- 关闭等温线； 0x01- 区间内模式 HH HH / LL LL：S16，高低温阈值，分辨率 0.1°C
指点测温	0x33	0x02 TT X0 X0 Y0 Y0	成功：0x00 失败：0x02	TT：U8，0x00- 关闭指点测温； 0x01- 开启指点测温 X0 X0 / Y0 Y0 : U16， [0,10000]，分别为光标测温目标点 的水平与垂直坐标，原点为图像左上角，X 坐标向右为正，Y 坐标向下为正，左上角坐 标为 [0,0], 右下角坐标为 [10000,10000]
OSD	0x73	TT	成功：0x00 失败：0x01	TT：U8，0x00-OSD 关； 0x01-OSD 开
画中画	0x74	TT	成功：0x00 失败：0x01	TT：U8，[0,4], 要切换的画中画模式。此值为 0 时，画中画模式循环切换
目标识别	0x75	TT	成功：0x00 失败：0x01	TT：U8，0x00- 目标识别关； 0x01- 目标识别开
变焦相机数字变焦	0x76	TT	成功：0x00 失败：0x01	TT：U8，0x00- 数字变焦关； 0x01- 数字变焦开
补光	0x80	TT	成功：0x00 失败：0x01	TT：U8，[0,255]，补光照明亮度
连续测距	0x81	TT	成功：0x00 失败：0x01	TT：U8，0x00- 测距关闭； 0x02- 测距开启

开启补光会同步开启夜视模式，关闭补光不会关闭夜视模式。

CRC 校验函数

```
uint16_t CalculateCrc16(uint8_t *ptr,uint8_t len)
{
        uint16_t crc;
        uint8_t da;
        uint16_t crc_ta[16]={
        0x0000,0x1021,0x2042,0x3063,0x4084,0x50a5,0x60c6,0x70e7,
        0x8108,0x9129,0xa14a,0xb16b,0xc18c,0xd1ad,0xe1ce,0xf1ef,
        };
        crc=0;
        while(len--!=0)
        {
                da=crc>>12;
                crc<<=4;
                crc^=crc_ta[da^(*ptr>>4)];
                da=crc>>12;
                crc<<=4;
                crc^=crc_ta[da^(*ptr&0x0F)];
                ptr++;
        }
        return(crc);
}
```

附录 1 上位机发送数据包转换示例

字节	内容	原始数据	转换精度 或转为二进制	转为 16 进制 （小端序）
0	协议头	0xA8	-	A8
1		0xE5	-	E5
2-3	包长度	72	72	48 00
4	协议版本	V0.2	-	02
5-6	滚转控制量	0	0	00 00
7-8	俯仰控制量	100	100	64 00
9-10	偏航控制量	-100	-100	9C FF
11	状态标志	控制量有效 载机惯导数据有效	0000 0101	05
12-13	载机绝对滚转角	-11.3213°	-1132	94 FB
14-15	载机绝对俯仰角	1.01°	101	65 00
16-17	载机绝对偏航角	240°	24000	C0 5D
18-19	载机北向加速度	1.123m/s^2	112	70 00
20-21	载机东向加速度	-1.123m/s^2	-112	90 FF
22-23	载机天向加速度	1.123m/s^2	112	70 00
24-25	载机北向速度	21.123m/s	2112	40 08
26-27	载机东向速度	-21.123m/s	-2112	C0 F7
28-29	载机天向速度	21.123m/s	2112	40 08
30	GCU 返回数据副帧请求码	0x01	-	01
31-36	预留	-	-	00 00 00 00 00 00
37	副帧帧头	0x01	-	01
38-41	载机经度	170.917533212	1709175332	24 F2 DF 65
42-45	载机纬度	38.030082231	380300822	16 EE AA 16
46-49	载机海拔	41.1231m	41123	A3 A0 00 00
50	载机卫星数	19	19	13
51-54	载机 GNSS 毫秒数	352718000	352718000	B0 0C 06 15

字节	内容	原始数据	转换精度或转为二进制	转为 16 进制（小端序）
55-56	载机 GNSS 周数	2278	2278	E6 08
57-60	相对高度	12.12m	12120	58 2F 00 00
61-68	预留	-	-	00 00 00 00 00 00 00 00
69	空命令	0x00	-	00
70-71	CRC	-	-	5D 1B

上位机发送的完整数据包为：

A8 E5 48 00 02 00 00 64 00 9C FF 05 94 FB 65 00 C0 5D 70 00 90 FF 70 00 40 08 C0 F7 40 08 01 00 00 00 00 00 00 01 24 F2 DF 65 16 EE AA 16 A3 A0 00 00 13 B0 0C 06 15 E6 08 58 2F 00 00 00 00 00 00 00 00 00 00 00 5D 1B

附录 2 GCU 返回数据包转换示例

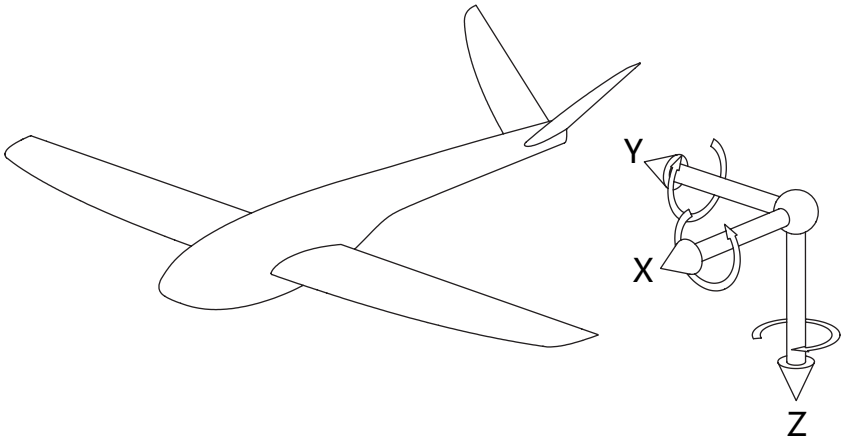
GCU 返回的完整数据包为：

8A 5E 49 00 02 12 01 80 0C FE F4 01 DD FC 20 00 4A 18 FF FF A5 03 47 18 FF FF 01 00 FE FF 00 00 00 00 00 00 00 01 1F 32 29 00 00 06 17 00 00 24 F2 DF 65 16 EE AA 16 A3 A0 00 00 2B 01 14 00 00 00 00 08 00 00 20 00 EC 85

字节	内容	原始数据（16 进制）	解析数据
0	协议头	A8	0xA8
1		5E	0x5E
2~3	包长度	49 00	73
4	版本号	02	V0.2
5	吊舱工作模式	12	指向跟随
6~7	吊舱状态标志	01 80	0000 0001 1000 0000 测距开 测距值与目标点坐标有效
8~9	水平方向脱靶量	0C FE	-500
10~11	垂直方向脱靶量	F4 01	500
12~13	相机 X 轴相对角度	DD FC	-8.03°
14~15	相机 Y 轴相对角度	20 00	0.32°
16~17	相机 Z 轴相对角度	4A 18	62.18°
18~19	相机绝对滚转角	FF FF	-0.01°
20~21	相机绝对俯仰角	A5 03	9.33°
22~23	相机绝对偏航角	47 18	62.15°
24~25	相机 X 轴绝对角速度	FF FF	-0.1deg/s
26~27	相机 Y 轴绝对角速度	01 00	0.1deg/s
28~29	相机 Z 轴绝对角速度	FE FF	-0.2deg/s
30~36	预留	00 00 00 00 00 00 00	-
37	副帧帧头	01	-
38	硬件版本	1F	3.1
39	固件版本	32	5.0
40	吊舱代码	29	D-90AI
41~42	故障代码	00 00	-
43~46	目标距离	06 17 00 00	589.4m
47~50	目标经度	24 F2 DF 65	170.9175332
51~55	目标纬度	16 EE AA 16	38.0300822
55~58	目标海拔高度	A3 A0 00 00	41.123m
59~60	1 号相机当前倍率	2B 01	29.9x

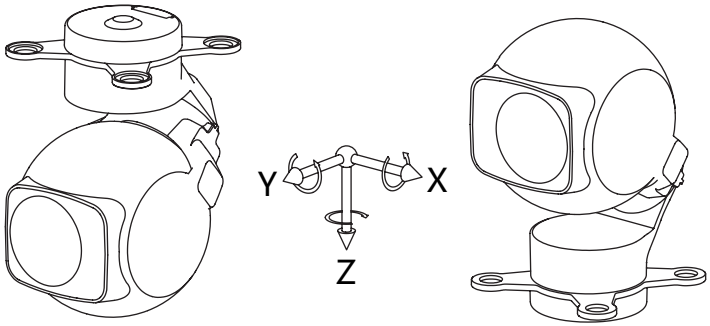
字节	内容	原始数据（16 进制）	解析数据
61~62	2 号相机当前倍率	14 00	2x
63	热成像相机状态标志	00	-
64~65	相机状态标志	00 00	-
66	时区	08	utc+8
67~68	预留	00 00	-
69~70	命令反馈	20 00	命令反馈：拍照执行成功
71~72	CRC	EC 85	-

附录 3 载机坐标系定义



附录 4 相机坐标系定义及旋转顺序

1. 坐标系定义：



正置上电　　倒置上电

吊舱控制接口需指向载机 X 轴负方向，减震平台需与载机 XOY 平面平行，吊舱安装时请尽量靠近载机质心

2. 旋转顺序：Z → Y → X。

3. 角度转换：

定义：

CamPhi：相机绝对滚转角（GCU 返回数据主帧 18~19 字节）
CamThe：相机绝对俯仰角（GCU 返回数据主帧 20~21 字节）
CamPsi：相机绝对指向角（GCU 返回数据主帧 22~23 字节）
AngleX：相机 X 轴绝对角度
AngleY：相机 Y 轴绝对角度
AngleZ：相机 Z 轴绝对角度

以上参数经过如下变换：

```
AngleZ += 90;
WARP (AngleZ , 360);
CamPhi = +AngleY;
CamThe = -AngleX;
CamPsi = +AngleZ;
```

附录 5 示例数据包

空命令
A8 E5 48 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FD 13

俯仰控制（保持当前控制模式，控制量 100）
A8 E5 48 00 02 00 00 64 00 00 00 04 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 E7 9F

俯仰控制（保持当前控制模式，控制量 -100）
A8 E5 48 00 02 00 00 9C FF 00 00 04 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 0F 0E

指向控制（保持当前控制模式，控制量 1000）
A8 E5 48 00 02 00 00 00 00 E8 03 04 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 DC 69

回中
A8 E5 48 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 03 CD 70

OSD 载机坐标
A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 06 00 E7 0D

OSD 目标点坐标
A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 06 01 F7 2C

自动翻转开

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 07 00 D4 3C

自动翻转关

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 07 01 C4 1D

时区设置（UTC-2）

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 08 FE CA D3

角度控制（控制量无效）

A8 E5 48 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 10 01 86

角度控制（欧拉角：滚转 0°，俯仰 45°，指向 60°）

A8 E5 48 00 02 00 00 94 11 70 17 04 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 10 4A 53

角度控制（欧拉角：滚转 20°，俯仰 0°，指向 0°）

A8 E5 48 00 02 D0 07 00 00 00 00 04 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 10 05 F7

锁定（控制量无效）

A8 E5 48 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 11 FF 03

指向锁定（指向相对角速度 +10° /s）
A8 E5 48 00 02 00 00 00 00 E8 03 04 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 11 DE 79

指向跟随（控制量无效）
A8 E5 48 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 12 CF 60

俯拍（控制量无效）
A8 E5 48 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 13 DF 41

欧拉角控制（控制量无效）
A8 E5 48 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 14 41 02

欧拉角控制（欧拉角：滚转 0°，俯仰 -45°，指向 0°）
A8 E5 48 00 02 00 00 6C EE 00 00 04 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 14 A5 6A

进入跟踪（X0=100，Y0=100，X1=105，Y1=105）
A8 E5 52 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 17 01 01 64 00 64 00 69 00 69 00 20 55

退出跟踪
A8 E5 52 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 17 01 00 64 00 64 00 69 00 69 00 CB 76

指点平移（X=100，Y=100）

A8 E5 4D 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 1A 01 64 00 64 00 48 AF

指点平移（X=5000，Y=5000）

A8 E5 4D 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 1A 01 88 13 88 13 EE C9

指点平移（X=10000，Y=10000）

A8 E5 4D 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 1A 01 10 27 10 27 53 65

指点平移（X=10000，Y=5000）

A8 E5 4D 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 1A 01 10 27 88 13 B4 F0

FPV（控制量无效）

A8 E5 48 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 1C 2E AE

拍照

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 20 01 5B 6C

录像开始 / 停止

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 21 01 68 5D

1 号相机连续放大

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 22 01 3D 0E

1 号相机连续缩小

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 23 01 0E 3F

1 号相机停止变倍

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 24 01 97 A8

指定 1 号相机倍率 (5000, 对应最大倍率的一半)

A8 E5 4B 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 25 01 88 13 73 0C

指定所有相机倍率 (1.0x)

A8 E5 4B 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 25 FF F6 FF F6 21

指定所有相机倍率 (5.5x)

A8 E5 4B 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 25 FF C9 FF E3 8A

指定 1 号相机倍率 (-603，(60.3x))
A8 E5 4B 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 25 01 A5 FD 75 DC

聚焦
A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 26 01 F1 CA

调色盘循环切换
A8 E5 4A 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 2A 02 00 8B E6

调色盘切换至模式 3
A8 E5 4A 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 2A 02 03 BB 85

夜视开
A8 E5 4A 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 2B 01 01 F9 A4

夜视关
A8 E5 4A 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 2B 01 00 E9 85

区域测温开 (X0=4000，Y0=4000，X1=6000，Y1=6000)
A8 E5 52 00 02 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 30 02 01 A0 0F A0 0F 70 17 70 17 BE 6D

区域测温关

A8 E5 52 00 02 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 30 02 00 00 00 00 00 00 00 00 00 00 33 96

温度报警开（高温警告温度 30.2° C，低温警告温度 20.0° C）

A8 E5 4E 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 31 02 01 2E 01 C8 00 E8 93

温度报警关

A8 E5 4E 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 31 02 00 2E 01 C8 00 42 C2

等温线开（区间内模式，15.0° C~25.2° C）

A8 E5 4E 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 32 02 01 FC 00 96 00 6F FA

等温线关

A8 E5 4E 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 32 02 00 FC 00 96 00 C5 AB

指点测温开（X=4000，Y=5000）

A8 E5 4E 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 33 02 01 A0 0F 88 13 C3 8D

指点测温关

A8 E5 4E 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 33 02 00 00 00 88 13 AF 9B

OSD 开

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 73 01 B8 60

OSD 关

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 73 00 A8 41

画中画循环切换

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 74 00 89 36

画中画切换至模式 3

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 74 03 B9 55

目标识别开

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 75 01 AA 26

目标识别关

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 75 00 BA 07

变焦相机数字变焦开

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 76 01 FF 75

变焦相机数字变焦关

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 76 00 EF 54

补光开（亮度 255）

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 80 FF 48 C3

补光关

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 80 00 56 33

连续测距开

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 81 02 45 40

连续测距关

A8 E5 49 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 81 00 65 02

附录 6 GPS 时间与 UTC 转换函数（未处理闰秒）

```
static const uint16_t gpst0[] = {1980, 1, 6, 0, 0, 0};
uint64_t epoch2time(const uint16_t *ep)
{
        const uint16_t _day[] = {1, 32, 60, 91, 121, 152, 182, 213, 244, 274, 305,
        335};
        uint64_t seconds = 0;
        uint16_t days, year = ep[0], mon = ep[1], day = ep[2];
        if (year < 1970 || 2099 < year || mon < 1 || 12 < mon) return seconds;
        /* leap year if year%4==0 in 1901-2099 */
        days=(year-1970)*365+(year-1969)/4+_day[mon-1]+day-2+(year%4==0
        && mon>=3?1:0);
        seconds = floor(ep[5]);
        seconds = (uint64_t)days * 86400 + ep[3] * 3600 + ep[4] * 60 + seconds;
        return seconds;
}
uint64_t gpst2time(int16_t week, uint32_t sec)
{
        uint64_t t = epoch2time(gpst0);
        if (sec < -1E9 || 1E9 < sec) sec = 0.0;
        t += 86400 * 7 * week + sec;
        return t;
}
uint8_t time2gps(uint64_t time, int16_t *week, uint32_t *msec)
{
        uint64_t t = epoch2time(gpst0);
        t = time - t;
        *week = t / 604800; // 604800=7*86400
        *msec = (t % 604800) * 1000;
        return 1;
}
```

附录 7 吊舱型号代码

吊舱代码	吊舱型号
0	Z-6A
2	Z-6C
25	Z-8RB
26	Z-8RC
31	Z-9B_V3
40	D-80AI
41	D-90AI
44	D-80Pro
45	D-90Pro(TA)
49	Z-1Pro
50	Z-1Mini
52	Z-2Mini
53	D-125AI(T)
55	D-90DE
57	D-125AI(V)
58	Z-9B_V4(T)
59	Z-9B_V4(V)
60	D-90Pro(VA)
61	D-90Pro(T)
62	D-90Pro(V)

(T)：测温型　　(V)：观瞄型

(TA)：测温型 + AICore　　(VA)：观瞄型 + AICore

附录 8 控制命令支持列表

表中未列控制命令为全系列机型支持。

命令	Z-6A	Z-6C	Z-8RB	Z-8RC	Z-9B_V3	D-80AI
跟踪	●	○	●	○	●	●
OSD 坐标	○	○	○	○	○	○
图像自动翻转	○	○	○	○	○	○
时区	○	○	○	○	○	○
指定倍率	○	○	○	○	○	●
聚焦	●	●	●	●	●	●
调色盘	○	○	○	○	●	○
夜视	●	●	●	●	●	●
区域测温	○	○	○	○	○	○
温度报警	○	○	○	○	○	○
等温线	○	○	○	○	○	○
指点测温	○	○	○	○	○	○
OSD	○	○	○	○	○	●
画中画	○	○	○	○	●	●
目标识别	○	○	○	○	○	○
数字变焦	○	○	○	○	○	○
补光	○	○	○	○	●	●
连续测距	○	○	●	●	●	○

命令	D-90AI	D-80Pro	D-90Pro(TA)	Z-1Pro	Z-1Mini	Z-2Mini
跟踪	●	●	●	●	●	●
OSD 坐标	○	○	●	●	●	●
图像自动翻转	○	○	●	●	●	●
时区	○	○	●	●	●	●
指定倍率	●	●	●	●	●	●
聚焦	●	●	●	○	○	○
调色盘	●	○	●	○	○	●
夜视	●	●	●	○	○	○
区域测温	○	○	●	○	○	○
温度报警	○	○	●	○	○	○
等温线	○	○	●	○	○	○
指点测温	○	○	●	○	○	○
OSD	●	○	●	●	●	●
画中画	●	○	●	○	○	●
目标识别	○	○	●	●	●	●
数字变焦	○	○	●	○	○	○
补光	○	●	○	○	○	○
连续测距	●	○	●	○	○	○

命令	D-125AI(T)	D-90DE	D-125AI(V)	Z-9B_V4(T)	Z-9B_V4(V)
跟踪	●	●	●	●	●
OSD 坐标	●	○	●	●	●
图像自动翻转	●	○	●	●	●
时区	●	○	●	●	●
指定倍率	●	●	●	●	●
聚焦	●	●	●	●	●
调色盘	●	○	●	●	●
夜视	●	●	●	●	●
区域测温	●	○	○	●	○
温度报警	●	○	○	●	○
等温线	●	○	○	●	○
指点测温	●	○	○	●	○
OSD	●	●	●	●	●
画中画	●	●	●	●	●
目标识别	●	○	●	●	●
数字变焦	●	○	●	●	●
补光	○	○	○	●	●
连续测距	●	●	●	●	●

命令	D-90Pro(VA)	D-90Pro(T)	D-90Pro(V)
跟踪	●	○	○
OSD 坐标	●	●	●
图像自动翻转	●	●	●
时区	●	●	●
指定倍率	●	●	●
聚焦	●	●	●
调色盘	●	●	●
夜视	●	●	●
区域测温	○	●	○
温度报警	○	●	○
等温线	○	●	○
指点测温	○	●	○
OSD	●	●	●
画中画	●	●	●
目标识别	●	○	○
数字变焦	●	●	●
补光	○	○	○
连续测距	●	●	●